

Instructions

- Submit the assignments in time via online form [here](#).
- The solutions MUST be done in $\text{\LaTeX}$ and handed as pdf. No handwritten assignments will be accepted.
- For python based problems, provide the codes as well as any generated figure. Seek your instructor's or peers' help if you need it. Responses during the viva that amount to: '...that's what the AI gave me', would result in a 0.
- Equations must be numbered and properly formatted.
- Expect some simple questions from the readings/viewings in the exam as well.
- Your final marks in the assignment will be contingent on the viva.

$\text{\LaTeX}$ Tips

If you feel your $\text{\LaTeX}$ file is getting unwieldy, you can use the `\input` command to divide it into smaller parts. See [this](#).

Vector Calculus

1. Solve problem 1.62 of Griffiths.
2. Solve problem 1.48 of Griffiths

Electrostatics

3. A long non-magnetic cylindrical conductor with inner radius a and outer radius b carries a current I . The current density in the conductor is uniform. Find the magnetic field set up by this current as a function of radius
 - a) inside the hollow space ($r < a$);
 - b) within the conductor ($a < r < b$);
 - c) outside the conductor ($r > b$).

Python: Plot the field of an electric dipole using python and superimpose the electric potential lines on top of it. See images for [reference](#).

Magnetostatics

4. A long non-magnetic cylindrical conductor with inner radius a and outer radius b carries a current I . The current density in the conductor is uniform. Find the magnetic field set up by this current as a function of radius
 - a) inside the hollow space ($r < a$);
 - b) within the conductor ($a < r < b$);
 - c) outside the conductor ($r > b$).

Python: Plot the field you derived as a function of the r .

Python

5. Consider the Gaussian function

$$f(x) = A \exp \left[-\frac{(x - \mu)^2}{2\sigma^2} \right].$$

Write a Python program that performs the following tasks:

- (a) **Define the Gaussian function.**

Define a Python **function** for $f(x)$ with A , μ , and σ as parameters.

- (b) **Plot the Gaussian.**

Generate an array of x -values over the interval

$$\mu - 5\sigma \leq x \leq \mu + 5\sigma$$

and use `matplotlib` to plot the Gaussian function.

(c) **Calculate the integral numerically.**

Numerically evaluate

$$I = \int_{\mu-5\sigma}^{\mu+5\sigma} f(x) dx.$$

(d) **Investigate the effect of σ .**

Repeat the calculation for at least three different values of σ , and plot the corresponding Gaussian functions **on the same graph**.

Use

$$A = 1, \quad \mu = 0, \quad \sigma = 0.5, 1, 2.$$

(e) **Compare with the analytical result.**

The exact integral of the Gaussian over the entire real line is

$$\int_{-\infty}^{\infty} f(x) dx = A\sigma\sqrt{2\pi}.$$

Compare your numerical result with this analytical value and calculate the percentage error:

$$\text{Percentage Error} = \left| \frac{I_{\text{numerical}} - I_{\text{analytical}}}{I_{\text{analytical}}} \right| \times 100\%.$$

Present your results in a table.

(f) **Discussion.**

Briefly answer the following questions:

- i. What happens to the width of the Gaussian as σ increases?
- ii. What happens to the maximum value of the Gaussian when A is kept constant?
- iii. How does the area under the curve change with σ ?
- iv. Why does integrating from $\mu - 5\sigma$ to $\mu + 5\sigma$ provide a good approximation to the integral over the entire real line?

Requirements

Your program should use:

- `numpy` for numerical calculations and arrays;
- `matplotlib` for plotting;
- a user-defined Python function for the Gaussian;
- a numerical integration method available in Python.

Your submission should include the Python code, the resulting plot, a table of numerical and analytical integrals, and brief answers to the discussion questions.

Readings

Pendry, Schurig, and Smith. *Controlling Electromagnetic Fields*. See Readings folder.